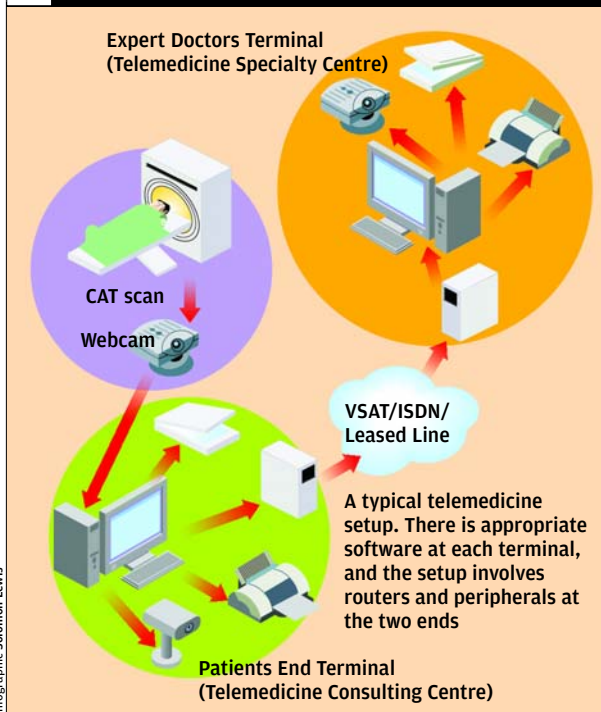


Telemedicine How The Setup Works

Infographic: Solomon Lewis



icine links with Colombo and Dhaka. The hospital plans to introduce telemedicine links with Nepal, Kazakhstan and Bhutan.

The letters pour in: one from Aragonda, a remote village in Andhra Pradesh, reads: "My daughter F B, three and a half years old, was operated for a hole in her heart at Chennai. We used to travel to Chennai every six months for checkups. Now, through telemedicine, we can consult the same doctor, from Aragonda, instead of going all the way to Chennai."

Telemedicine does away with distance barriers when it comes to healthcare. It saves time and money for those who are lucky enough to be able to use it. So what exactly is telemedicine, and what does it involve?

The What And The How

Essentially, telemedicine involves sharing and exchanging of patient related data and medical opinions between a medical specialist and a doctor in a remote location through telecommunication networks. Telemedicine thus overcomes geographical barriers, can make for quality and effective healthcare, and is eminently cost-effective.

Apart from the basic idea—that telemedicine is used to reach out to patients who do not have proper

healthcare facilities—what are the applications? There are several, as Dr D Lavanian, Business Manager—Telemedicine, Apollo Telemedicine Networking Foundation, Apollo Hospitals Group, tells us:

- ▣ Relief efforts during natural calamities
- ▣ Continuing medical education through videoconferencing
- ▣ Providing training to prevent/respond to typical disease patterns in remote areas
- ▣ Adding to healthcare knowledge bases: telemedicine can be a communication tool to aid in the understanding of the nature of ailments
- ▣ In case of emergencies, a general practitioner can perform surgery with tele-monitored specialist assistance
- ▣ In second opinions, where the diagnosis can be confirmed and treatment planned

Objectives And Offerings

Apollo aims at nothing less than reaching the masses in India where healthcare is not available, under the belief that every citizen is entitled to advanced and quality medical care. They aim to set up a referral system as well. General practitioners (GPs) are targeted under their program: Apollo hopes to develop and demonstrate telemedicine for use by GPs. And, Apollo strives to evaluate the impact of telemedicine in terms of economy and user satisfaction.

Apollo's offerings include consulting for telemedicine projects; telemedicine software; IT hardware; medical hardware;

hardware and software for security; communication links; and operational management and remote support.

The Aragonda Project

Situated a little distance from Tirupati in Andhra Pradesh, Aragonda and six neighbouring villages were part of a pilot project by Apollo. The area is agricultural, and consists of a 7,000 people in a 1,500 families. Aragonda had very few facilities with only six general practitioners, and one poly-health clinic—30 kilometres away. Apollo set up a 50-bed hospital with state-of-the-art telemedicine infrastructure in January of 2000. ISRO provided the VSAT equipment and free connectivity. This was India's first rural telemedicine centre, and has helped significantly in improving the health status of the population. It is a self-sustaining model, with each family paying a rupee a day—which covers medical treatment up to Rs 20,000.

An Example Treatment

At the GB Pant Hospital in Port Blair lay an unconscious patient. Admitted following a head injury, the nature of the wound was uncertain. Photographs of the pupil of the patient were taken. These were sent to Prof K Ganapathy, Senior Consultant Neurosurgeon at Madras, along with the CT scans. The scans showed primary injury to the brain, but it turned out that the patient did not require surgery. The local doctor treated him with the assistance of a teleconsultant. The patient was discharged in just 10 days.

Telemedicine works

The bottomline is, is telemedicine effective? Is it worth the investment? Can it actually save lives? The statistics speak for themselves: from January 2000 to June 2002, out of 3,986 consultations in all Apollo's telemedicine centres, 3,395 of them—or 85 per cent—were successfully treated at the telemedicine centres. Only 15 per cent had to be referred to specialty centres.

All it takes for telemedicine to become big is some bandwidth, some collaboration, and a vision like Apollo's. ☐

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A live telemedicine consultation between Aragonda and Hyderabad

